

Lab Report: EEE 4878

Group No. #10

Lab No. #04

LAB TITLE:

**Design and Simulation of DC and AC
Characteristics of a Two-Stage CMOS
Operational Amplifier Using Cadence**

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ABSTRACT

This laboratory experiment presents the design and simulation of an unbuffered two-stage CMOS operational amplifier using Cadence Virtuoso. The operational amplifier consists of an NMOS differential input pair, a PMOS active-load network, a common-source second stage, and a frequency-compensation capacitor.

The DC transfer characteristic was investigated by sweeping the non-inverting input voltage from -2 V to $+2\text{ V}$ while maintaining the inverting input voltage at 0 V . The resulting transfer characteristic was used to determine the input and output offset voltages. Small-signal AC analysis was performed from 100 Hz to 1 GHz to investigate the low-frequency voltage gain, dominant-pole frequency, gain-bandwidth product, phase margin, and gain margin.

The positive and negative power-supply rejection ratios were evaluated using transfer-function analysis. The common-mode rejection ratio was also simulated after modifying the operational-amplifier circuit to include external positive and negative supply pins. Finally, the width of the output PMOS transistor was adjusted to reduce the original offset voltage of approximately 609.8 mV .

The experiment demonstrated the effects of transistor dimensions, frequency compensation, supply disturbances, and common-mode input signals on the performance of a two-stage CMOS operational amplifier.

KEYWORDS

1. OpAmp
2. CMOS
3. Cadence
4. PSRR
5. CMRR

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1 INTRODUCTION

An operational amplifier is a high-gain electronic circuit that amplifies the difference between two input voltages. Operational amplifiers are widely used in signal amplification, active filters, feedback systems, oscillators, voltage regulators, data converters, and instrumentation circuits.

This experiment focuses on the design and simulation of an unbuffered two-stage CMOS operational amplifier. The first stage consists of an NMOS differential input pair with a PMOS active load, while the second stage is a common-source voltage-amplification stage.

The experiment was expected to provide an understanding of the DC transfer characteristic, input and output offset voltages, frequency response, dominant pole, gain-bandwidth product, phase margin, gain margin, power-supply rejection ratio, and common-mode rejection ratio.

These parameters are important in practical analog circuit design. Low offset voltage improves accuracy, sufficient phase margin ensures stable feedback operation, high PSRR reduces the effect of power-supply noise, and high CMRR enables the amplifier to reject unwanted common-mode signals.

2 THEORY

2.1 Operational Amplifier

The output voltage of an operational amplifier can be expressed as

$$V_{\text{out}} = A_v (V_+ - V_-), \quad (1)$$

where A_v is the open-loop voltage gain, V_+ is the non-inverting input voltage, and V_- is the inverting input voltage.

The differential input voltage is

$$V_{id} = V_+ - V_-. \quad (2)$$

2.2 Two-Stage CMOS Operational Amplifier

The first stage of the designed operational amplifier is an NMOS differential amplifier with a PMOS active load. This stage converts the differential input voltage into a single-ended output signal.

The second stage is a common-source amplifier that provides additional voltage gain. The total low-frequency gain can be approximated as

$$A_v \approx A_{v1}A_{v2}, \quad (3)$$

where A_{v1} and A_{v2} are the voltage gains of the first and second stages, respectively.

A compensation capacitor is connected between the first-stage output and the final output node. This capacitor creates a dominant pole and improves the stability of the amplifier.

2.3 DC Transfer Characteristic

The DC transfer characteristic represents the relationship between the output voltage and the differential input voltage.

The DC gain near the high-gain transition region can be estimated using

$$A_{v,DC} = \frac{\Delta V_{out}}{\Delta V_{in}}. \quad (4)$$

2.4 Output Offset Voltage

The output offset voltage is the output voltage obtained when the two input voltages are equal:

$$V_{OS,out} = V_{out} \Big|_{V_+ = V_-}. \quad (5)$$

2.5 Input Offset Voltage

The input offset voltage is the differential input voltage required to make the output voltage equal to zero:

$$V_{OS,in} = (V_+ - V_-) \Big|_{V_{out}=0}. \quad (6)$$

2.6 Frequency Response

The voltage gain in decibels is

$$A_v(\text{dB}) = 20 \log_{10} \left| \frac{V_{out}}{V_{in}} \right|. \quad (7)$$

The dominant pole is the lowest-frequency pole that causes the gain to start decreasing. For an approximately single-pole amplifier, the gain-bandwidth product is

$$GBW \approx A_0 f_p, \quad (8)$$

where A_0 is the low-frequency gain in linear form and f_p is the dominant-pole frequency.

2.7 Phase Margin

The phase margin is determined at the unity-gain frequency:

$$PM = 180^\circ + \angle A(j\omega_u), \quad (9)$$

where ω_u is the unity-gain angular frequency.

2.8 Gain Margin

The gain margin is measured at the frequency where the phase shift reaches -180° :

$$GM_{\text{dB}} = -20 \log_{10} |A(j\omega_{180})|. \quad (10)$$

2.9 Power Supply Rejection Ratio

The positive power-supply rejection ratio is

$$PSRR^+ = \left| \frac{V_{DD}}{V_{\text{out}}} \right|. \quad (11)$$

The negative power-supply rejection ratio is

$$PSRR^- = \left| \frac{V_{SS}}{V_{\text{out}}} \right|. \quad (12)$$

In decibels,

$$PSRR_{\text{dB}} = 20 \log_{10}(PSRR). \quad (13)$$

2.10 Common-Mode Rejection Ratio

The common-mode rejection ratio is defined as

$$CMRR = \left| \frac{A_d}{A_{cm}} \right|, \quad (14)$$

where A_d is the differential gain and A_{cm} is the common-mode gain.

For the unity-gain test circuit used in this experiment,

$$CMRR = \left| \frac{V_{cm}}{V_{out}} \right|. \quad (15)$$

In decibels,

$$CMRR_{dB} = 20 \log_{10}(CMRR). \quad (16)$$

3 PRE-WORK DATA AND RESULTS

3.1 Component Specifications

The initial transistor dimensions are listed in Table 1.

Table 1: Transistor dimensions of the two-stage CMOS operational amplifier

Transistor	Type	Length	Width
M_1	PMOS	$1 \mu\text{m}$	$94 \mu\text{m}$
M_2	PMOS	$1 \mu\text{m}$	$15 \mu\text{m}$
M_3	PMOS	$1 \mu\text{m}$	$15 \mu\text{m}$
M_4	NMOS	$1 \mu\text{m}$	$4.5 \mu\text{m}$
M_5	NMOS	$1 \mu\text{m}$	$4.5 \mu\text{m}$
M_6	NMOS	$1 \mu\text{m}$	$3 \mu\text{m}$
M_7	NMOS	$1 \mu\text{m}$	$3 \mu\text{m}$
M_8	NMOS	$1 \mu\text{m}$	$14 \mu\text{m}$

Table 2: Voltage-source, current-source, and capacitor specifications

Parameter	Value
Positive supply voltage, V_{DD}	+2.5 V
Negative supply voltage, V_{SS}	-2.5 V
Bias-current source	30 μ A
Compensation capacitor, C_1	5 pF
Load capacitor, C_2	10 pF
V_+ DC voltage	inp
V_+ AC magnitude	1 V
V_- DC voltage	0 V
V_- AC magnitude	0 V

3.2 MOSFET Model File

The following Level-2 transistor models were saved in a file named `corner.scs`.

```
simulator lang=spice

* VTI-derived Level=2 nominal model

.model nmos4 nmos level=2 vto=0.775 gamma=0.4
+ tox=400e-10 nsub=8e+15 xj=0.15U ld=0.20U
+ u0=650 ucrit=0.62e+5 uexp=0.125
+ vmax=5.1e+4 neff=4.0 delta=1.4 rsh=36
+ cgso=1.95e-10 cgdo=1.95e-10
+ cj=195U cjsw=500P mj=0.76 mjsw=0.30 pb=0.8

.model pmos4 pmos level=2 vto=-0.75 gamma=0.57
+ tox=400e-10 nsub=6e+15 xj=0.05U ld=0.20U
+ u0=255 ucrit=0.86e+5 uexp=0.29
+ vmax=3.0e+4 neff=2.65 delta=1.0 rsh=101
+ cgso=1.90e-10 cgdo=1.90e-10
+ cj=250U cjsw=350P mj=0.535 mjsw=0.34 pb=0.8
```

4 LAB HANDOUT QUESTIONS

4.1 Question 1

Provide the circuit diagrams and performance curves of the circuits.

Answer:

The original operational-amplifier schematic, DC transfer curve, frequency-response graph, PSRR circuit and curves, modified CMRR circuit, CMRR response, updated PMOS circuit, and corrected transfer curve are presented in this report.

4.2 Question 2

Answer all the questions given in Task 4.2.

4.3 Question 3

The initial offset voltage is approximately 609.8 mV. Adjust the width of the output PMOS transistor to bring the offset voltage close to zero. Show the updated circuit and transfer characteristic.

5 TOOLS USED

1. Cadence Virtuoso Schematic Editor
2. Cadence Virtuoso Analog Design Environment L
3. Spectre Circuit Simulator
4. Virtuoso Visualization and Analysis Calculator
5. `analogLib` component library
6. Linux terminal
7. Text editor for creating the `corner.scs` model file

6 PROCEDURE

1. A new schematic cell view was created in Cadence Virtuoso.
2. The `pmos4` and `nmos4` transistors were selected from the `analogLib` library.
3. The transistor lengths and widths were entered according to the supplied operational-amplifier circuit.
4. Two DC voltage sources were connected to provide +2.5 V and -2.5 V supply voltages.
5. A 30 μ A current source was used to establish the reference bias current.

6. A 5 pF compensation capacitor and a 10 pF load capacitor were connected.
7. Three pins named $V+$, $V-$, and V_{out} were created.
8. The circuit was checked using the *Check and Save* command.
9. The MOSFET model definitions were saved in the `corner.scs` file.
10. The model file was added through the Model Library Setup option in ADE L.
11. For DC analysis, the non-inverting input source was assigned the variable `inp`.
12. The variable `inp` was swept from -2 V to $+2$ V.
13. The inverting input source was maintained at 0 V.
14. The output voltage was plotted against the swept input voltage.
15. Markers were placed on the transfer characteristic to determine the input and output offset voltages.
16. For AC analysis, the sweep frequency was set from 100 Hz to 1 GHz.
17. The output-voltage magnitude was converted to decibels using the `dB20` function.
18. The dominant-pole frequency and unity-gain frequency were identified from the gain response.
19. The `gainBWProd`, `PhaseMargin`, and `GainMargin` functions were selected in the Virtuoso Calculator.
20. A three-pin operational-amplifier symbol was created for the PSRR simulation.
21. Transfer-function analysis was configured from 10 Hz to 0.01 GHz.
22. The positive supply source was selected to obtain the $PSRR^+$ response.
23. The negative supply source was selected to obtain the $PSRR^-$ response.
24. For the CMRR simulation, the internal VDD and VSS reference components were removed.
25. External positive and negative supply pins were added.
26. A new five-pin operational-amplifier symbol was generated.
27. The differential and common-mode test circuits were constructed.
28. AC analysis was performed from 10 Hz to 2.5×10^7 Hz.
29. The following expression was entered into the Virtuoso Calculator:

```
dB20(v("/Vout2" ?result "ac") /
      v("/Vout1" ?result "ac"))
```

30. The output PMOS transistor width was varied to minimize the offset voltage.
31. The updated circuit and corrected transfer characteristic were recorded.

7 RESULTS

7.1 Completed Operational-Amplifier Schematic

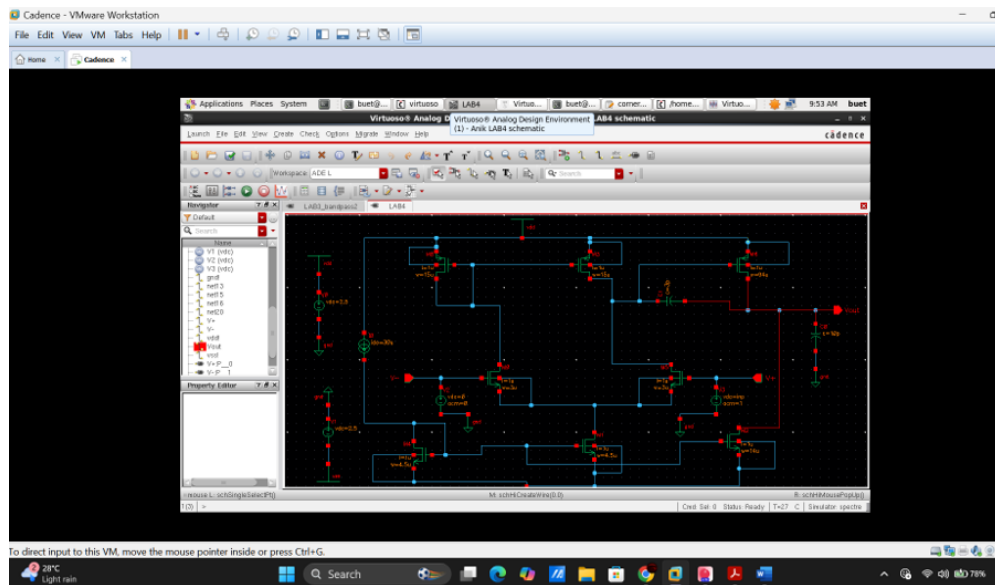


Figure 1: Completed schematic of the two-stage CMOS operational amplifier

The completed circuit contained an NMOS differential input pair, a PMOS active-load current mirror, NMOS bias-current sources, a common-source second stage, a Miller compensation capacitor, and an output load capacitor.

7.2 DC Transfer Characteristic

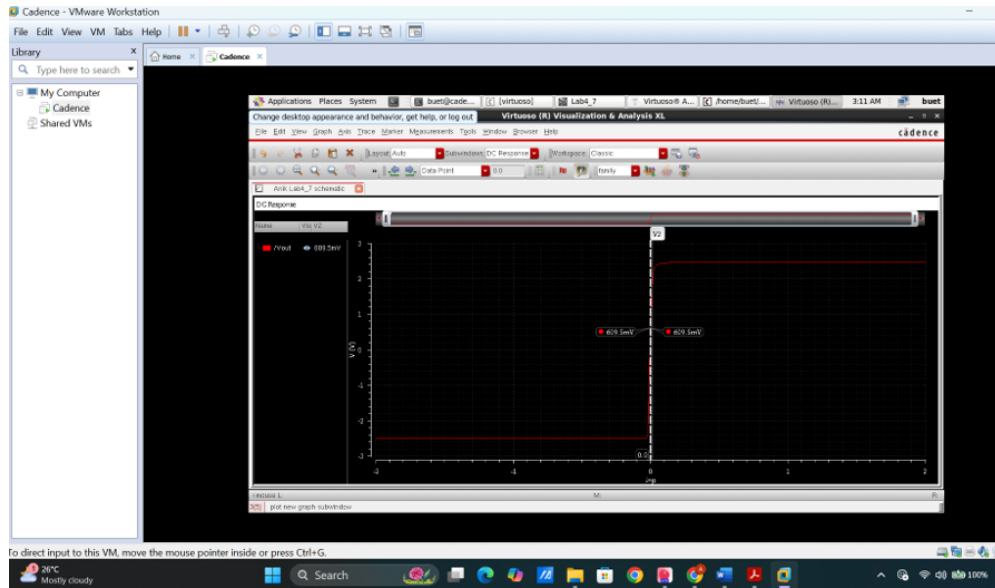


Figure 2: DC transfer characteristic of the two-stage CMOS operational amplifier

The output offset voltage measured at $V_+ = V_- = 0$ was

$$V_{OS,out} = 609.5\text{mV}. \quad (17)$$

7.3 Frequency-Response Results



Figure 3: Frequency Response of the two-stage CMOS operational amplifier

The AC response produced the following approximate results:

Table 3: Parameters determined from the AC response

Parameter	Approximate value
Dominant-pole frequency	38 kHz
Gain-bandwidth product	5.7 MHz
Estimated phase margin	2°
Gain margin within displayed range	Greater than 55 dB

7.4 Power-Supply Rejection Ratio Simulation

The Power-Supply Rejection Ratio (PSRR) indicates the ability of the operational amplifier to prevent variations in the supply voltages from appearing at the output.

The positive and negative power-supply rejection ratios are defined as

$$PSRR^+ = \left| \frac{V_{DD}}{V_{out}} \right| \quad (18)$$

and

$$PSRR^- = \left| \frac{V_{SS}}{V_{out}} \right|. \quad (19)$$

The corresponding values in decibels are calculated using

$$PSRR_{dB} = 20 \log_{10}(PSRR). \quad (20)$$

The operational-amplifier symbol was connected in a feedback configuration to perform the PSRR simulation. The transfer-function analysis was performed over a frequency range from 10 Hz to 0.01 GHz.

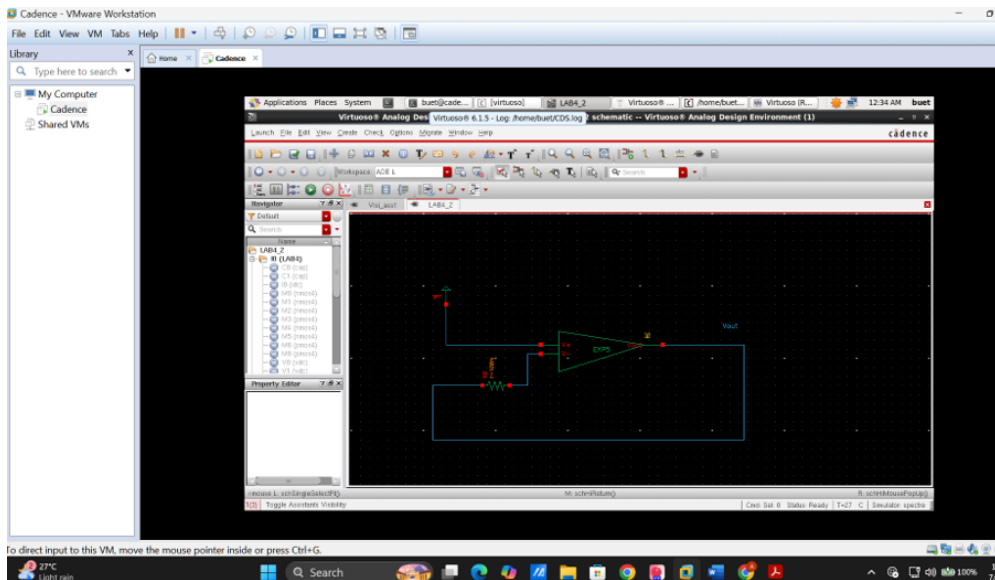


Figure 4: Schematic used for the simulation of PSRR

For the simulation of $PSRR^+$, the positive output node was selected as V_{out} , while the internal positive supply node V_{DD} was selected as the negative output node in the transfer-function analysis.

For the simulation of $PSRR^-$, the positive output node remained V_{out} , while the internal negative supply node V_{SS} was selected as the negative output node.

Positive Power-Supply Rejection Ratio

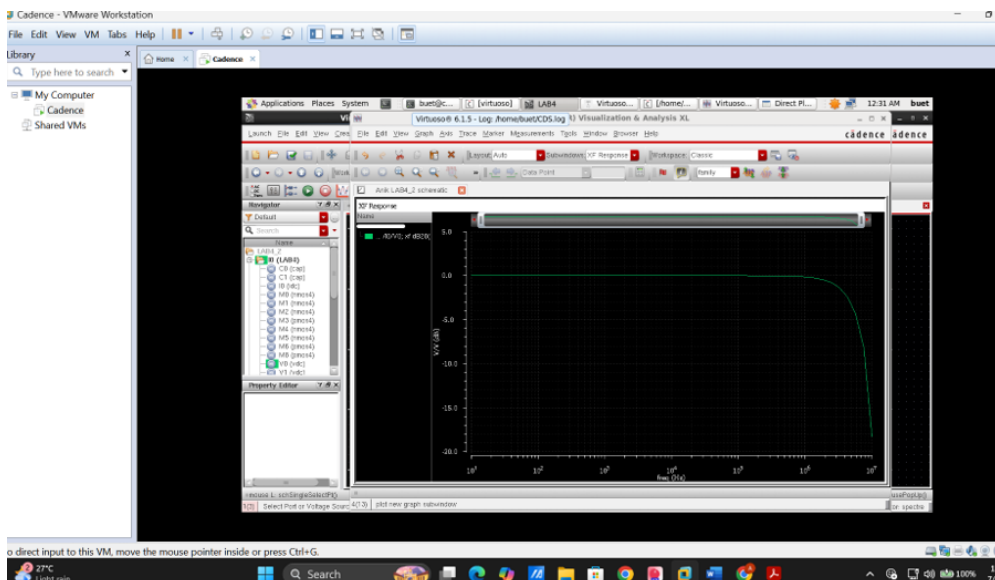


Figure 5: Frequency response of the positive power-supply rejection ratio

Negative Power-Supply Rejection Ratio

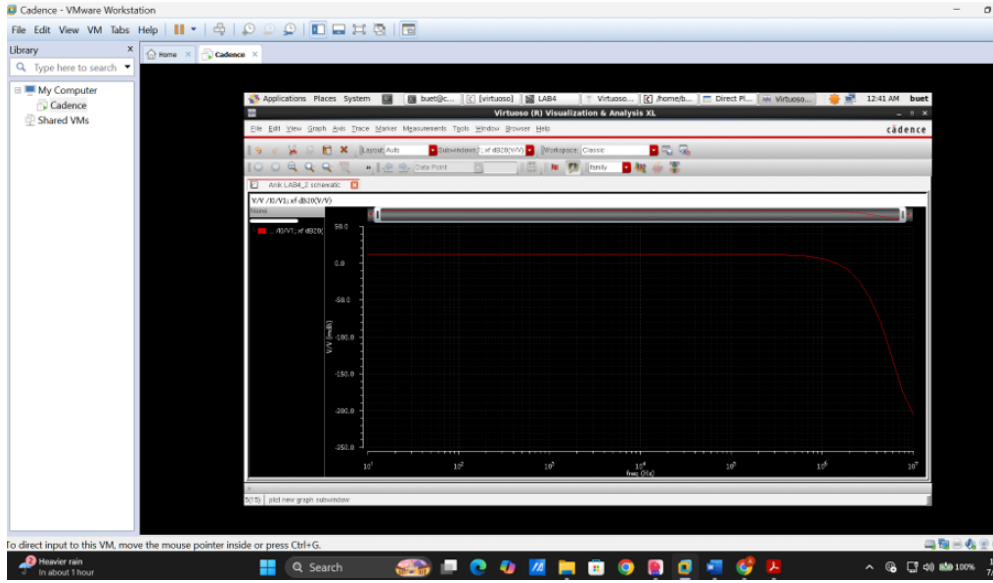


Figure 6: Frequency response of the negative power-supply rejection ratio

7.5 Common-Mode Rejection Ratio Simulation

The Common-Mode Rejection Ratio (CMRR) represents the ability of the operational amplifier to reject signals that are applied equally to both input terminals.

The CMRR is defined as

$$CMRR = \left| \frac{A_d}{A_{cm}} \right|, \quad (21)$$

where A_d is the differential-mode gain and A_{cm} is the common-mode gain.

For the test configuration used in this experiment, the CMRR can be expressed as

$$CMRR = \left| \frac{V_{cm}}{V_{out}} \right|. \quad (22)$$

In decibels,

$$CMRR_{dB} = 20 \log_{10} \left| \frac{V_{cm}}{V_{out}} \right|. \quad (23)$$

To perform the CMRR simulation, the internal VDD and VSS reference components were removed from the operational-amplifier circuit. External positive and negative supply pins were then added so that the supply voltages could be connected from outside the operational-amplifier block.

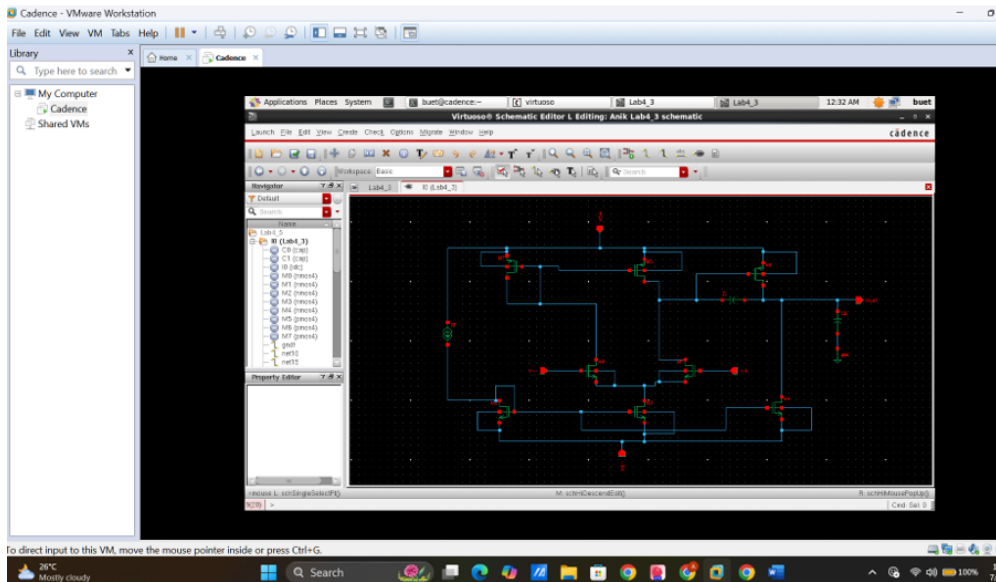


Figure 7: Modified OP AMP circuit for calculating CMRR

After modifying the internal circuit, a new operational-amplifier symbol was generated using the five pins $V+$, $V-$, V_{out} , V_{d+} , and V_{d-} . The modified symbol was then used to construct the CMRR simulation setup.

The common-mode and differential-response circuits were connected as shown in Figure 8. The two outputs were labelled V_{out1} and V_{out2} .

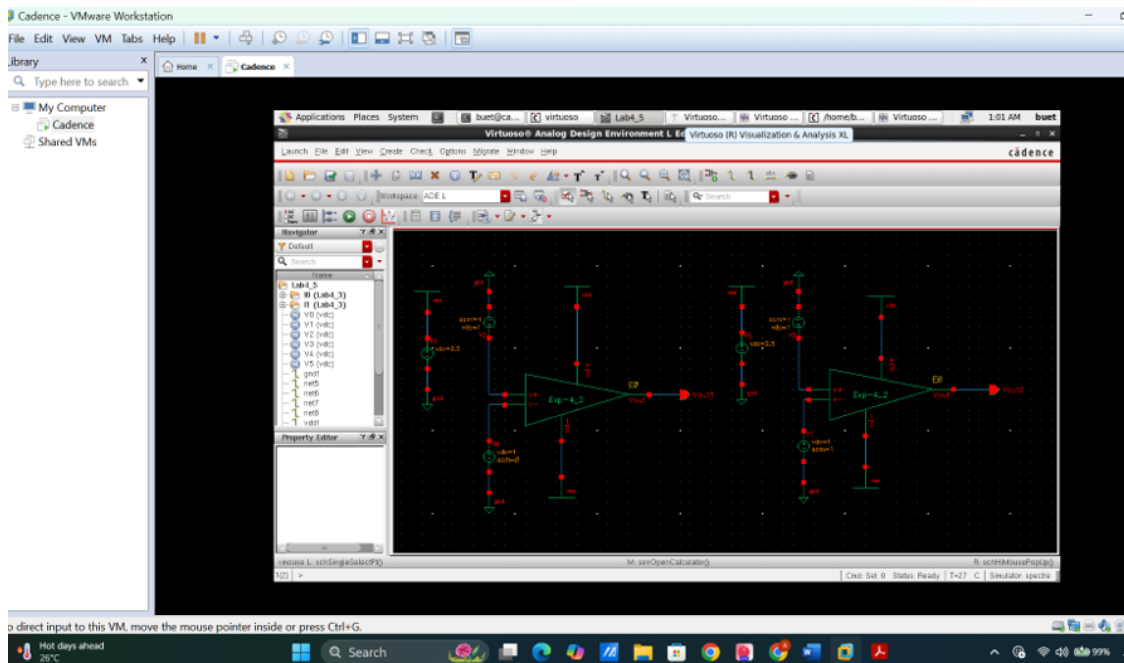


Figure 8: Set-up for the simulation of the CMRR

An AC analysis was performed over a frequency range from 10 Hz to

$$2.5 \times 10^7 \text{ Hz.} \quad (24)$$

The CMRR response was obtained from the ratio of the two output voltages:

$$CMRR_{dB} = 20 \log_{10} \left| \frac{V_{out2}}{V_{out1}} \right|. \quad (25)$$

The following expression was entered into the Virtuoso Calculator:

```
dB20(v("/Vout2" ?result "ac") /
      v("/Vout1" ?result "ac"))
```

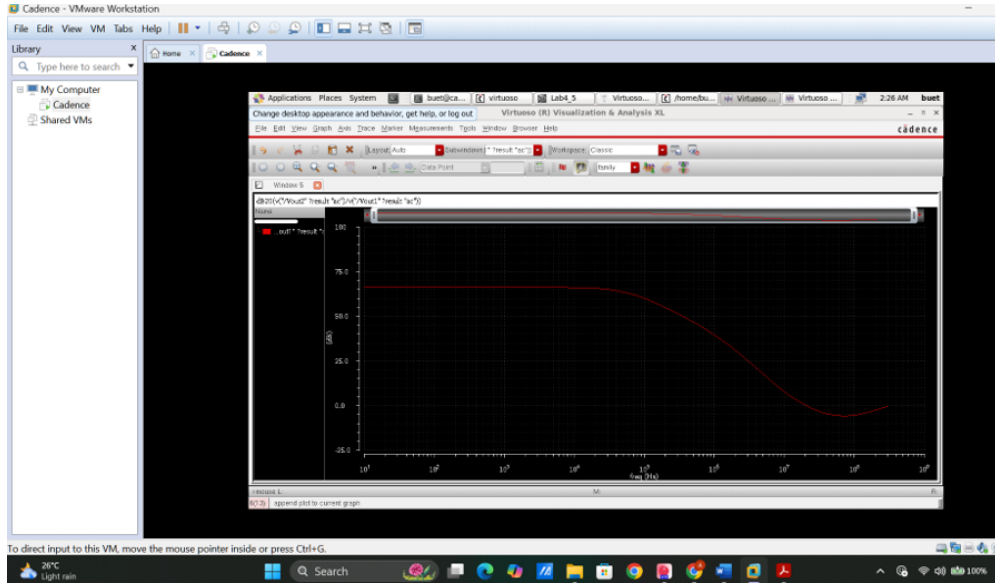


Figure 9: Frequency response of CMRR

7.6 Offset Voltage Reduction by Adjusting the Width of Output PMOS M8

The initial DC transfer characteristic showed an offset voltage of approximately 609.8 mV. To minimize this offset voltage, the width of the output PMOS transistor M_8 was varied while keeping the remaining transistor dimensions and circuit parameters unchanged.

The original width of the output PMOS transistor was

$$W_{M8, \text{original}} = 94 \mu\text{m}. \quad (26)$$

The updated operational-amplifier circuit used for this analysis is shown in Figure 10.

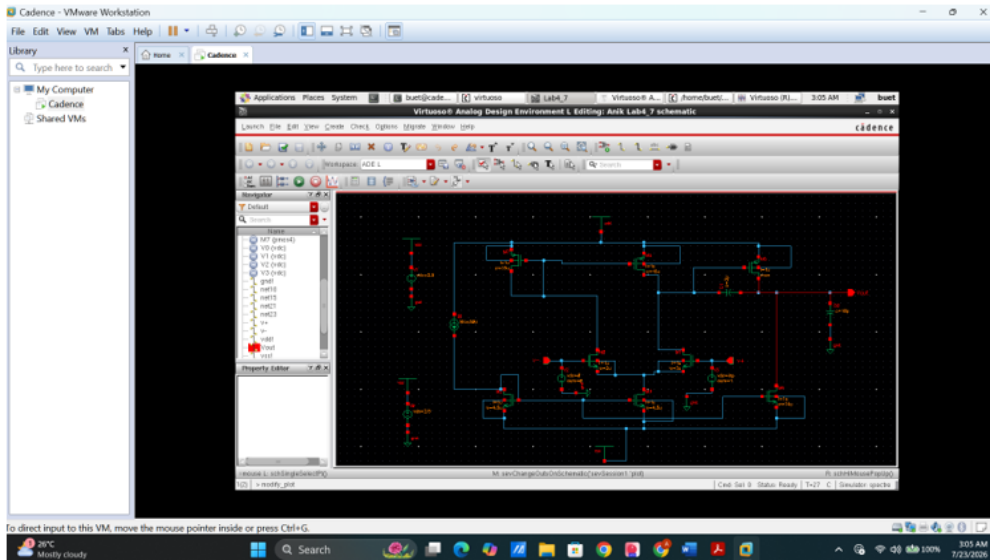


Figure 10: Updated operational-amplifier circuit for adjusting the width of output PMOS M_8

A parametric analysis was performed by defining the width of transistor M_8 as the design variable w . The width was swept linearly from

$$72.10 \mu\text{m} \quad (27)$$

to

$$72.11 \mu\text{m} \quad (28)$$

using a total of 50 simulation steps.

The parametric analysis configuration and the corresponding transfer characteristics are shown in Figure 11.

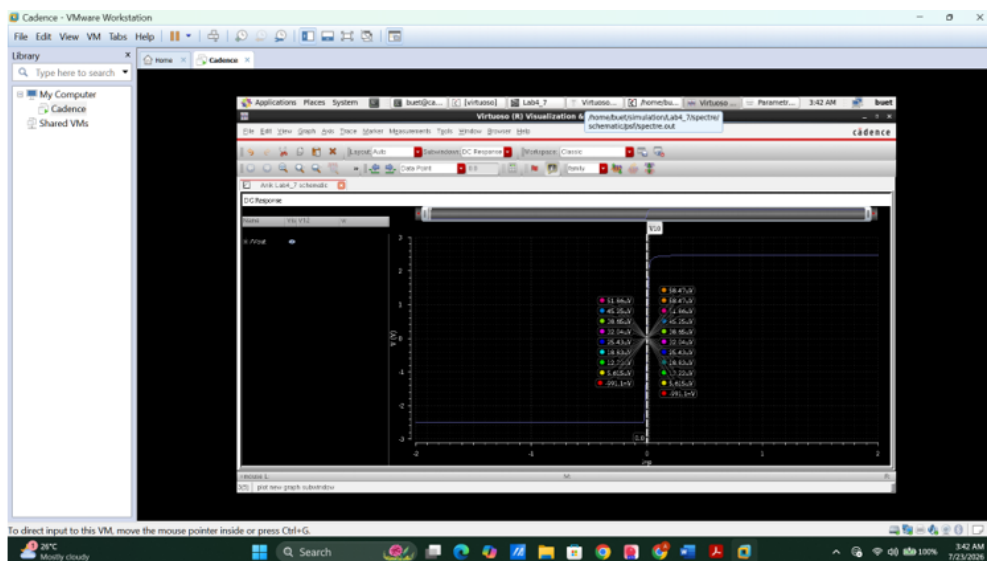


Figure 11: Parametric analysis of the width of output PMOS M_8

From the parametric analysis, the output offset voltage became closest to zero when the width of the output PMOS transistor was

$$W_{M_8, \text{modified}} = 72.10 \mu\text{m}. \quad (29)$$

After setting the width of M_8 to $72.10 \mu\text{m}$, the DC analysis was performed again by sweeping the input variable inp from -2 V to $+2 \text{ V}$.

The corrected transfer characteristic is shown in Figure 12.

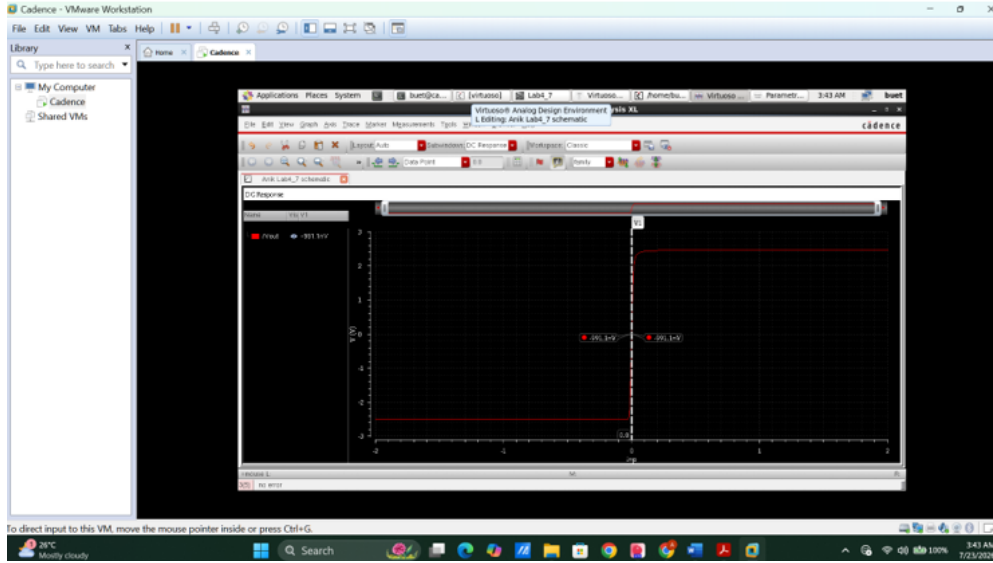


Figure 12: DC transfer characteristic after adjusting the width of output PMOS M_8

From the corrected transfer characteristic, when

$$\text{inp} = 0 \text{ V}, \quad (30)$$

the output voltage was approximately

$$V_{\text{out}} = -991 \text{ nV}. \quad (31)$$

Converting this value into microvolts gives

$$V_{\text{out}} = -0.991 \mu\text{V}. \quad (32)$$

In millivolts,

$$V_{\text{out}} = -0.000991 \text{ mV}. \quad (33)$$

Therefore, the corrected output offset voltage was

$$\boxed{V_{\text{OS,out,corrected}} \approx -991 \text{ nV} \approx 0 \text{ V}}. \quad (34)$$

The percentage reduction in the magnitude of the output offset voltage was calculated using

$$\% \text{ offset reduction} = \frac{|V_{\text{OS,out,initial}}| - |V_{\text{OS,out,corrected}}|}{|V_{\text{OS,out,initial}}|} \times 100\%. \quad (35)$$

Substituting the obtained values,

$$\% \text{ offset reduction} = \frac{609.8 - 0.000991}{609.8} \times 100\%. \quad (36)$$

Therefore,

$$\boxed{\% \text{ offset reduction} \approx 99.99984\%}. \quad (37)$$

Thus, reducing the width of output PMOS transistor M_8 from $94 \mu\text{m}$ to $72.10 \mu\text{m}$ significantly improved the balance between the pull-up and pull-down currents of the output stage. As a result, the output voltage at zero differential input was reduced from approximately 609.8 mV to only -991 nV , which can be considered practically zero.

8 CONCLUSION

A two-stage CMOS operational amplifier with an NMOS differential input pair was designed and simulated using Cadence Virtuoso. The DC transfer analysis demonstrated the high-gain transition characteristic of the amplifier and revealed a non-zero offset voltage in the original circuit.

The AC analysis showed that the low-frequency voltage gain was approximately 44 dB . The dominant-pole frequency was approximately 38 kHz , while the unity-gain frequency and gain-bandwidth product were approximately 5.7 MHz .

The estimated phase margin was approximately 2° , indicating poor relative stability. No clear phase crossover was observed within the displayed frequency range, and the gain was already approximately -55 dB at 1 GHz . Therefore, the gain margin was greater than approximately 55 dB within the simulated range.

The positive and negative power-supply rejection simulations showed how variations in the two supply voltages affected the output. The difference between $PSRR^+$ and $PSRR^-$ resulted from the different paths through which the positive and negative supply disturbances entered the circuit.

The CMRR analysis demonstrated the ability of the operational amplifier to reject common-

mode input signals. The CMRR decreased at high frequencies because of transistor parasitic capacitances and unequal internal signal paths.

The original offset voltage was approximately 609.8 mV. The output PMOS width was changed from $94\mu\text{m}$ to _____ μm , reducing the output offset voltage to _____ mV.

Overall, the experiment provided practical experience in CMOS operational-amplifier design, DC analysis, AC analysis, frequency compensation, stability evaluation, power-supply rejection, common-mode rejection, and offset-voltage minimization.

9 REFERENCES

- [1] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd ed., McGraw-Hill Education, 2016.
- [2] P. R. Gray, P. J. Hurst, S. H. Lewis, and R. G. Meyer, *Analysis and Design of Analog Integrated Circuits*, 5th ed., John Wiley & Sons, 2009.
- [3] Sheikh Munim Hussain and Liman Shams, "Design and Simulation of DC and AC Characteristics of a Two-Stage CMOS Operational Amplifier Using Cadence," EEE 4878 Analog Integrated Electronics Laboratory, Islamic University of Technology.
- [4] Cadence Design Systems, *Virtuoso Analog Design Environment and Spectre Circuit Simulator Documentation*.